

Pranav Cheraku

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EDUCATION

Western Washington University

Bellingham, WA

Bachelor of Science in Computer Science, Minor in Mathematics

Expected June 2026

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Systems, Machine Learning, Deep Learning, Operating Systems, Linear Algebra, Probability and Statistics, Multivariable Calculus

EXPERIENCE

Software Implementation Consultant

June 2025 – September 2025

FAST Enterprises LLC

Harrisburg, PA

- Delivered production updates to reports, letters, and batch jobs serving **6M+ Pennsylvania tax filers** by writing and deploying **VB.NET and SQL** changes in GenTax
- Reduced system maintenance overhead by retiring **110+ legacy jobs**, identified by tracing execution logic and database queries to flag jobs no longer in use
- Translated ambiguous tax compliance requirements into production logic for GenTax by investigating business rules with SMEs and validating implementations through code review and testing

Robotics Researcher

April 2025 – Present

MRRP Lab

Bellingham, WA

- Built a real-time **YOLO** object detection pipeline that locates and classifies target objects in real time for a **UR3e robotic arm**, replacing hardcoded target coordinates with live perception
- Designed a voice interface using a local **LLM** with speech-to-text and text-to-speech, letting users issue spoken commands to the robot during collaborative tasks
- Developed autonomous pick-and-place routines on the UR3e using **MoveIt** and **ROS**, planning safe paths to objects identified by the vision pipeline

Freelance Web Developer

May 2025 – July 2025

Olympia Hindu Temple & Cultural Center

Olympia, WA

- Designed and shipped the official website for the Olympia Hindu Temple & Cultural Center, a nonprofit serving **350+ community members**, replacing manual event communication with a centralized digital hub
- Built a responsive frontend with **HTML, JavaScript, and Tailwind CSS**, including a dynamic Panchang calendar that auto-updates daily without manual editing

PROJECTS

Fake Image Detector | *Python, PyTorch, Torchvision, NumPy, scikit-learn, Pillow*

March 2025

- Engineered a **GAN**-generated fake face classifier achieving **99.98% accuracy** with **zero false positives** on a 15,324-sample test set by designing a PyTorch **Discriminator** with residual attention and dual classification heads
- Reduced misclassifications to **3 of 15,324** by benchmarking 4+ **CNN** architectures and implementing ReduceLROnPlateau LR scheduling, gradient clipping, data augmentation, and hyperparameter tuning

Relational Database Management System | *Java, Relational Algebra, LinkedList, HashSet*

February 2025

- Engineered an in-memory **RDBMS** in **Java** with zero external dependencies, implementing a custom **lexical analyzer** and SQL-like DSL that parses both **DDL** and **DML** commands across hierarchical storage structures (HEAP, HASH, TREE) and a partitioned memory model
- Built a recursive query interpreter executing **relational algebra** operations (SELECT, PROJECT, JOIN) via **nested-loop theta joins**, handling 6 relational operators with AND/OR precedence in WHERE qualifiers and leveraging **secondary indices** for optimized multi-table queries

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, C, C++, C#, SQL

Frameworks & Libraries: FastAPI, Flask, React, Next.js, PostgreSQL, React Native, Tailwind CSS

Developer Tools: Git, Docker, AWS, CI/CD, Linux/Bash, ROS, REST APIs

AI/ML: PyTorch, TensorFlow, LLMs, RAG, Vector Databases

AWARDS

2nd Place at OSU BeaverHacks 2026 (Google Gemini Track) | 2nd Place at Insforge x Qoder AI Agent Hackathon 2026